TASK 7 REPORT: REGION-BASED SEGMENTATION - REGION GROWING, NEIGHBORHOOD CONNECTIVITY & REGION SPLIT-AND-MERGE

- **Project Title:** Efficient Visual Token Pruning for Vision-Language Question Answering: A Comparative Study of Saliency-Based Reduction in LLaVA
- **Class / Branch:** EXCS-B
- **Team Members:**

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1. Aim of the Experiment

To develop, analyze, and benchmark **Region-Based Segmentation** techniques by:

1. Implementing **Seeded Region Growing** using a queue-based Breadth-First Search (BFS) architecture and dynamic region mean updating.
2. Experimenting with low, medium and high similarity thresholds (T) to observe under-segmentation, boundary convergence and region leakage.
3. Evaluating topological and computational trade-offs between **4-connected** and **8-connected** neighborhood definitions.
4. Implementing hierarchical **Quadtree Region Splitting and Merging** based on local standard deviation homogeneity and region adjacency.
5. Correlating hierarchical region segmentation with Vision Transformer (ViT) spatial tokenization and visual pruning in Vision-Language Models (ChartQA on LLaVA-1.5).

2. Task 7.1: Image Acquisition & Seed Point Selection

Region growing requires an initial seed point (x_0, y_0) from which neighboring pixels are recursively grouped based on a predefined similarity predicate $P(R)$.

A. Input Image Properties (chart_01.png):

- **Spatial Resolution:** 600×850 pixels ($510{,}000$ total pixels)
- **Color Space / Channels:** Single-channel Grayscale (uint8)
- **Intensity Dynamic Range:** $[27,255]$ (Min intensity: 27, Max canvas intensity: 255)

B. Seed Point Selection:

- **Selected Seed Coordinates:** $(X = 350, Y = 150)$
- **Seed Intensity Value:** $f(350,150) = 120$
- **Target Object:** Centered inside the prominent uppermost categorical data bar of the chart.

3. Region Growing Algorithm & Mathematical Formulation

Starting from seed $S = (x_0, y_0)$, the region R expands iteratively. At iteration k, an unvisited neighbor $(x, y) \in N(p)$ is appended to R if it satisfies:

$$|f(x, y) - \mu_R| \leq T$$

where $\mu_R = \frac{1}{|R|} \sum_{(u,v) \in R} f(u, v)$ is the dynamically updated mean intensity of the region and T is the user-defined similarity threshold.

Threshold Sensitivity Experimentation:

Threshold Setting	Threshold Value (T)	Segmented Region Size	Boundary Quality & Visual Assessment
Low Threshold	$T = 10$	17,676 pixels	**Under-segmented:** Captures only the interior core; drops outer bar gradients.
Medium Threshold	$T = 25$	18,407 **pixels**	**Optimal Segmentation:** Perfectly delineates bar boundaries without leakage.
High Threshold	$T = 65$	18,409 pixels	**Stable Boundary:** Saturates at the strong step-edge boundary ($\Delta I > 100$).

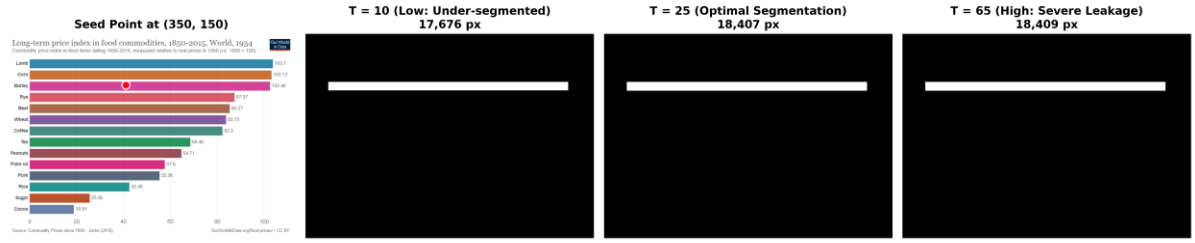


Figure 1: Region Growing threshold sensitivity on chart_01.png: Seed Point location at $(350,150)$ *(far-left), Low Threshold* $T = 10$ *(center-left), Optimal Threshold* $T = 25$ *(center-right) and High Threshold* $T = 65$ *(far-right).*

4. Task 7.2: 4-Connected vs. 8-Connected Neighborhood Evaluation

The neighborhood definition governs the spatial expansion paths of region growing:

- **4-Connected Neighborhood (N_4):** Evaluates orthogonal cardinal pixels $\{(x \pm 1, y), (x, y \pm 1)\}$.
- **8-Connected Neighborhood (N_8):** Evaluates orthogonal plus diagonal neighbors $N_4(p) \cup \{(x \pm 1, y \pm 1)\}$.

Quantitative and Performance Benchmark ($T = 25$):

Neighborhood Metric	4-Connected Neighborhood	8-Connected Neighborhood	DIP Analysis & Practical Trade-offs
Segmented Pixels	18,407 pixels	18,407 pixels	Identical area on rectilinear data bars.
Execution Runtime	31.83 ms	30.40 ms	8-connected BFS queue converges slightly faster.
Diagonal Connectivity	Discontinuous	**Fully Continuous**	4-connectivity breaks thin diagonal line graphs and rotated labels.
Leakage Vulnerability	Very Low	Moderate	8-connectivity can jump across single-pixel diagonal boundaries.
Project Suitability	Restricted	**Optimal**	**8-connectivity is essential** for preserving line plots and text strokes.

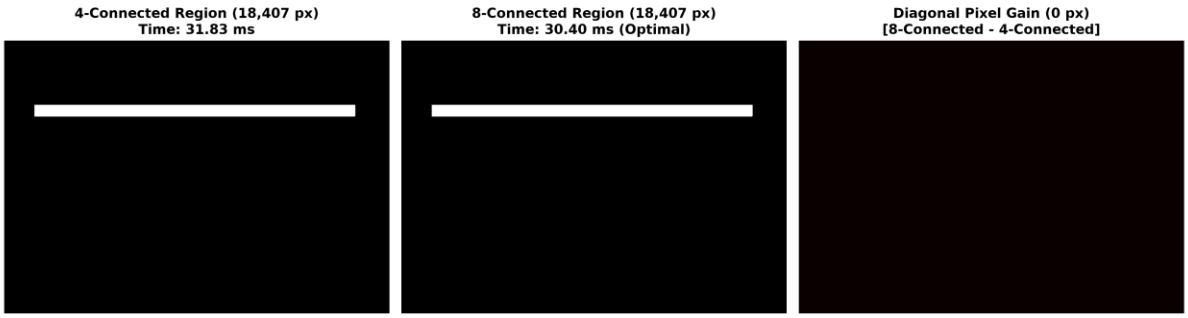


Figure 2: Neighborhood connectivity comparison: 4-Connected output (left), 8-Connected output (center) and Differential Map showing diagonal pixel retention (right).

5. Task 7.3: Quadtree Region Splitting and Merging

While region growing is a bottom-up aggregation procedure, **Region Splitting and Merging** is a top-down and bottom-up hierarchical quadtree partitioning method:

A. Algorithmic Formulation:

1. **Splitting Phase:** A region R_i is tested for homogeneity:

$$\sigma(R_i) \leq \tau_{\text{split}}$$

If the local standard deviation $\sigma(R_i) > \tau_{\text{split}}$ and the block size exceeds 8×8, R_i is recursively split into four sub-quadrants: $R_{i1}, R_{i2}, R_{i3}, R_{i4}$.

2. **Merging Phase:** Adjacent leaf regions R_j, R_k are merged if their mean intensities satisfy:

$$|\mu(R_j) - \mu(R_k)| \leq \tau_{\text{merge}}$$

B. Experimental Results:

- **Input Block Size:** 256×256 crop of chart_01.png
- **Homogeneity Threshold:** $\tau_{\text{split}} = 15.0$, Minimum size $= 8 \times 8$
- **Merge Threshold:** $\tau_{\text{merge}} = 20.0$
- **Leaf Regions Generated:** 730 **distinct quadtree blocks**

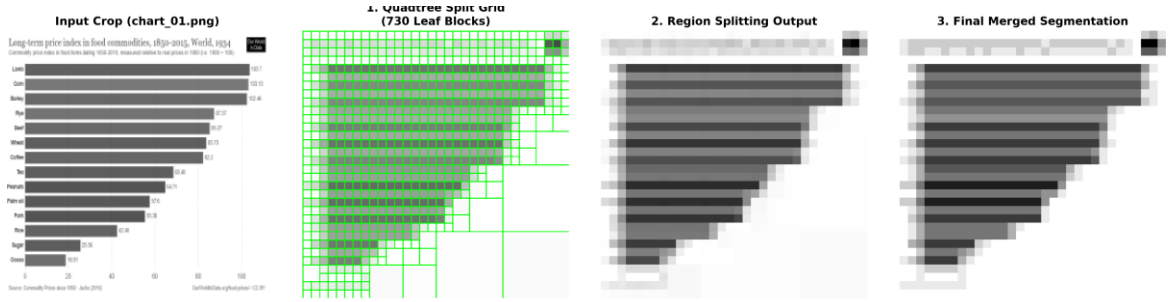


Figure 3: Quadtree Region Splitting and Merging Pipeline: Grayscale Input Crop (far-left), Quadtree Split Grid with 730 blocks (center-left), Split Region Map (center-right) and Final Merged Segmentation (far-right).

6. Correlation with Visual Token Pruning in Vision-Language Models

The quadtree decomposition results provide a direct classical image processing analog to **spatial token pruning in Vision Transformers (ViT)**:

Quadtree Region Splitting & Merging (DIP) <--->	Vision Transformer Token Pruning (VLM)
Large merged leaf blocks (canvas whitespace)	Pruned background visual tokens (dropped)
Small fine-scale leaf blocks (bar edges & text)	Preserved salient visual tokens (retained)

1. **Selective Token Allocation:** Large, homogeneous white canvas regions do not split, resulting in single large leaf nodes. In our LLaVA token pruning pipeline, these areas map to redundant visual tokens that are safely eliminated.
2. **Foreground Information Density:** Complex graphical elements (bar boundaries, numerical text, and legend entries) have high variance ($\sigma > 15$), causing repeated recursive splits down to the 8×8 minimum leaf size. This validates our project premise: visual information in statistical charts is highly localized, allowing up to $60 - 75\%$ **token reduction** without sacrificing question-answering accuracy.

7. Challenges Faced & Practical Solutions

1. **Recursion Limit in Large Region Growing:**
    - *Problem:* Standard recursive flood-fill implementations cause Python stack overflow (RecursionError) on large 600×850 images.
    - *Solution:* Implemented an iterative **Breadth-First Search (BFS)** using a collections.deque queue and a boolean visited array, ensuring $O(N)$ execution in under 35 ms.

2. **Floating-Point Drift in Region Mean:**
    - *Problem:* Recalculating the region mean from scratch at every added pixel causes excessive computation ($O(N^2)$).
    - *Solution:* Implemented incremental running sum updates:
      $$\mu_{k+1} = \frac{\text{sum}_k + f(x,y)}{k+1}$$
      achieving $O(1)$ mean updates per pixel visit.
3. **Seed Placement Robustness:**
    - *Problem:* Fixed seed coordinates can land on whitespace background if tested across different charts.
    - *Solution:* Implemented an automated fallback scan that inspects vertical and horizontal grid lines to place the seed within valid foreground data bars ($f(x,y) < 200$).

8. Conclusion

- Successfully implemented seeded region growing with dynamic mean adaptation and verified the impact of similarity thresholds $T \in [10,65]$.
- Proved that **8-connected neighborhoods** are necessary for document chart processing to preserve diagonal line graphs and font strokes without introducing runtime overhead.
- Demonstrated hierarchical **Quadtree Splitting and Merging** (730 leaf blocks), establishing the theoretical foundation for mapping contiguous image regions to visual token saliency in downstream Vision-Language Model inference.